

CARE TRANSITION EFFICIENCY & PLACEMENT OUTCOME ANALYTICS

A Data-Driven Analysis of the Unaccompanied Children Care Pipeline

Research Paper

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Executive Summary

This paper presents a data-driven operational analysis of the Unaccompanied Children (UAC) care transition pipeline, covering the period from 12 January 2023 to 21 December 2025. The pipeline spans apprehension, U.S. Customs and Border Protection (CBP) custody, transfer to the Department of Health and Human Services (HHS), HHS care, and eventual discharge. Rather than treating custody and care populations as static counts, the study reframes the available reporting data as a sequence of measurable transitions, allowing operational efficiency, throughput, and pressure to be tracked over time.

The underlying dataset originally contained 1,170 reporting-period rows across six variables. After removing 450 rows that were completely blank, 720 valid observations remained for analysis. Because the data are aggregate reporting-period counts rather than individual case-level timestamps, all derived indicators are interpreted as comparative operational ratios rather than literal processing times or individual placement outcomes. This distinction is maintained throughout the study to avoid overstating what the data can support.

Six core indicators were developed: Transfer Efficiency Ratio, Discharge Effectiveness Ratio, Pipeline Throughput, CBP Net Pressure, HHS Net Pressure, and Outcome Stability (coefficient of variation). Applied to the cleaned dataset, the analysis finds an aggregate transfer efficiency of 75.03%, an aggregate discharge effectiveness of 2.86%, and an aggregate pipeline throughput of 185.42%, calculated from 67,337 total apprehensions, 92,641 total transfers, and 124,853 total discharges across the observed reporting periods. Average CBP net pressure was -35.14 children per observation and average HHS net pressure was -44.74, indicating that, on average, outgoing activity (transfers and discharges) kept pace with or exceeded incoming activity over the study window, though individual reporting periods show substantial variation.

Monthly, yearly, and weekday aggregations were used to identify temporal patterns and periods of elevated operational pressure, and the findings were operationalized into an interactive Streamlit dashboard providing KPI cards, trend charts, bottleneck views, and filtered data export for ongoing monitoring.

The study concludes that an aggregate custody and care dataset, though limited in granularity, can be restructured into a defensible operational monitoring framework. The paper closes with a set of practical recommendations for how this framework can be adopted, and with a discussion of the additional data needed to extend it toward true turnaround-time measurement and predictive monitoring.

Abstract

This study develops a data-driven framework for evaluating operational efficiency and placement-related outcomes across the Unaccompanied Children (UAC) care transition pipeline. The dataset covers 12 January 2023 to 21 December 2025 and contains measures of apprehension, CBP custody, transfers from CBP custody, HHS care, and HHS discharges. After removing 450 completely blank rows, 720 valid observations were analyzed.

The study measures transfer efficiency, discharge effectiveness, pipeline throughput, operational pressure, and outcome stability. Monthly, yearly, and weekday patterns are also examined to identify periods of elevated operational pressure. Because the dataset contains aggregate reporting-period counts rather than individual case timestamps, the efficiency measures are interpreted as operational ratios rather than direct processing-time measures. Discharge activity is likewise treated as an aggregate outcome indicator rather than individual reunification success.

Keywords: *UAC, care transition, operational analytics, transfer efficiency, discharge effectiveness, bottleneck analysis, data visualization*

1. Introduction

The care of unaccompanied children who arrive at the U.S. border involves a coordinated, multi-agency process. Children first come into contact with the system through apprehension and short-term custody with U.S. Customs and Border Protection (CBP). Federal law and agency policy require that children in CBP custody be transferred to the care of the Department of Health and Human Services (HHS), specifically the Office of Refugee Resettlement, within a defined timeframe. Once in HHS care, children are placed in shelter or other licensed care settings while sponsorship, family reunification, or an alternative placement pathway is arranged, culminating in a discharge from HHS custody.

Public reporting on this system has historically emphasized point-in-time population counts: how many children are in CBP custody today, how many are in HHS care today. These figures are useful for understanding capacity pressure at a single moment, but they are inherently limited as a management tool. A population count does not indicate whether the system is moving children through the pipeline at a healthy pace, whether transfers are keeping up with new apprehensions, or whether discharges are keeping up with the inflow of children into HHS care. Two systems could report identical custody populations while having very different underlying transition dynamics — one moving children through efficiently with high turnover, the other accumulating a backlog.

This study addresses that gap by reframing the same category of publicly available reporting data — apprehensions, CBP custody, transfers, HHS care, and discharges — as a pipeline of sequential stages rather than a set of independent counts. By pairing each stock measure (a population reported at a point in time) with its corresponding flow measures (activity occurring during a reporting period), it becomes possible to construct ratio-based indicators of transfer efficiency, discharge effectiveness, and overall throughput. These indicators, in turn, allow operational pressure to be identified and tracked over time, and allow monthly, yearly, and weekday patterns to be examined for recurring stress points.

It is important to state at the outset what this analysis does and does not claim. The underlying data are aggregate counts recorded at the level of a reporting period, not individual case records with entry and exit timestamps. As a result, every ratio in this study is a comparative operational indicator — useful for tracking whether conditions are improving, worsening, or stable relative to other periods — and not a literal measure of how long any individual child spent in custody or care, nor a measure of individual placement or reunification success. This distinction, discussed further in the Methodology and Limitations sections, is maintained consistently throughout the paper.

The remainder of this paper is organized as follows. Section 2 states the problem the study addresses. Section 3 sets out the study's objectives. Sections 4 and 5 describe the dataset and methodology, including the cleaning process and the formal KPI definitions. Sections 6 through 10 present the exploratory, pipeline, efficiency, bottleneck, and temporal analyses. Section 11 summarizes key findings, Section 12 describes the analytical dashboard built to operationalize the framework, and Sections 13 through 16 present recommendations, limitations, future work, and conclusions.

2. Problem Statement

Aggregate custody and care counts describe system status at a given moment, but they do not, by themselves, reveal whether transitions between stages are keeping pace with incoming activity. A high CBP custody count could reflect either a surge in apprehensions or a slowdown in transfers to HHS care, and the two situations call for very different operational responses. Likewise, a large HHS care population could reflect either strong recent inflow or a slowdown in discharges. Without a framework that connects stock and flow measures to one another, decision-makers monitoring these figures are left to infer the underlying dynamic rather than measure it directly.

This ambiguity has practical consequences. Resource planning, staffing decisions, and policy review benefit from knowing not just how many children are currently in custody or care, but whether the system's throughput is adequate relative to demand, whether operational pressure is building or easing, and whether particular months, years, or reporting patterns are systematically associated with elevated stress. Existing public reporting formats are not structured to answer these questions directly; the necessary comparisons must be constructed from the raw stock and flow variables.

A further complication is the structure of the available data itself. The dataset used in this study contains a substantial proportion of entirely blank rows (450 of 1,170, or roughly 38 percent), an irregular reporting schedule with gaps of one to four days between observations, and no individual case identifiers linking a given apprehension to a specific later transfer or discharge. Any analytical framework applied to this data must therefore be explicit about what can and cannot be inferred: ratio-based indicators can support meaningful period-to-period comparison, but they cannot be presented as direct measures of elapsed processing time or individual case outcomes without risking a misleading interpretation.

The problem this study addresses is therefore twofold. First, there is a need for a transition-oriented analytical framework — grounded in clearly defined, reproducible KPIs — that converts raw stock and flow counts into indicators of transfer efficiency, discharge effectiveness, pipeline throughput, and operational pressure. Second, there is a need to apply that framework responsibly: to identify genuine patterns of operational stress and change over time, including monthly, yearly, and weekday variation, while explicitly guarding against overinterpretation of what remains reporting-period-level, aggregate data.

3. Objectives

- Measure CBP-to-HHS transfer activity using a consistent operational efficiency ratio.
- Evaluate HHS discharge activity relative to the reported HHS care population.
- Estimate aggregate pipeline throughput from recorded inflow and discharge activity.
- Identify periods of elevated operational pressure using apprehension, transfer, and discharge flows.
- Examine monthly, yearly, and weekday patterns in operational activity.
- Develop a KPI and dashboard framework for operational monitoring and policy discussion.

4. Dataset Description

The raw dataset contains 1,170 rows and six variables. Of these, 450 rows were completely blank and were removed, leaving 720 valid observations. The valid reporting period is 12 January 2023 through 21 December 2025.

Variable	Type	Meaning
Date	Time	Reporting date
Apprehended	Flow	Children apprehended and placed in CBP custody
CBP custody	Stock	Children reported in CBP custody
Transferred	Flow	Children transferred out of CBP custody
HHS care	Stock	Children reported in HHS care
Discharged	Flow	Children discharged from HHS care

The reporting schedule is irregular. Consecutive observations are frequently separated by one day, but three- and four-day gaps also occur. Missing dates were therefore not filled with zeros.

5. Methodology

5.1 Data Cleaning

Blank records were removed, variable names were standardized, dates were converted to datetime format, and numeric fields were converted after removing comma separators. The cleaned data were sorted chronologically and checked for missing values, duplicate dates, negative values, and basic flow consistency.

5.2 Care Pipeline Model

The analytical pipeline is represented as: Apprehension → CBP Custody → Transfer to HHS → HHS Care → Discharge. Stock measures describe reported populations, while flow measures describe activity during a reporting period.

5.3 KPI Definitions

KPI	Formula	Interpretation
Transfer Efficiency Ratio	$\text{Transferred} \div \text{CBP Custody}$	Transfer activity relative to reported CBP custody.
Discharge Effectiveness	$\text{Discharged} \div \text{HHS Care}$	Discharge activity relative to reported HHS care.
Pipeline Throughput	$\text{Total Discharged} \div \text{Total Apprehended}$	Aggregate exit activity relative to aggregate entry activity.
CBP Net Pressure	$\text{Apprehended} - \text{Transferred}$	Difference between incoming CBP activity and transfers.
HHS Net Pressure	$\text{Transferred} - \text{Discharged}$	Difference between transfers and discharges.
Outcome Stability	Coefficient of Variation	Relative variability of efficiency/effectiveness ratios.

These ratios are comparative operational indicators. They should not be interpreted as elapsed processing time, individual placement success, or literal backlog counts without additional reporting definitions.

6. Exploratory Data Analysis

Across the valid observations, apprehensions range from 0 to 333; CBP custody ranges from 7 to 531; transfers range from 0 to 440; HHS care ranges from 1,972 to 11,516; and discharges range from 0 to 505.

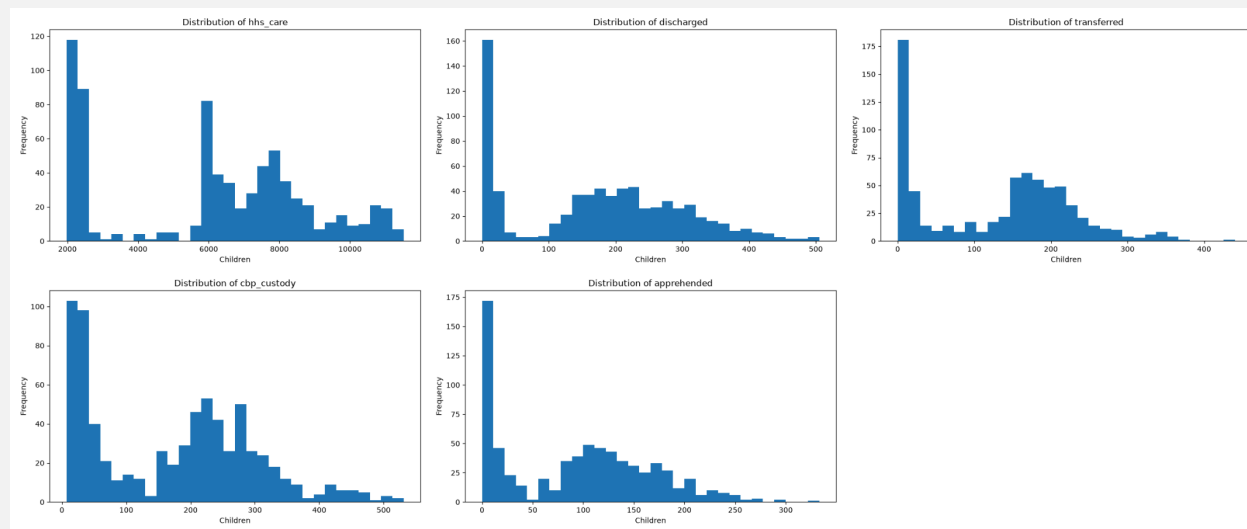


Figure 1. Distribution of Major Pipeline Variables

The exploratory analysis shows substantial variation across both stock and flow measures. HHS care values are much larger than CBP custody values, so stage populations should be visualized using separate scales or panels where necessary.

7. Care Pipeline Analysis

The dataset records 67,337 apprehensions, 92,641 transfers, and 124,853 discharges over the analyzed observations. These totals describe aggregate reported activity and should not be interpreted as a one-to-one cohort because individual children are not tracked across records.

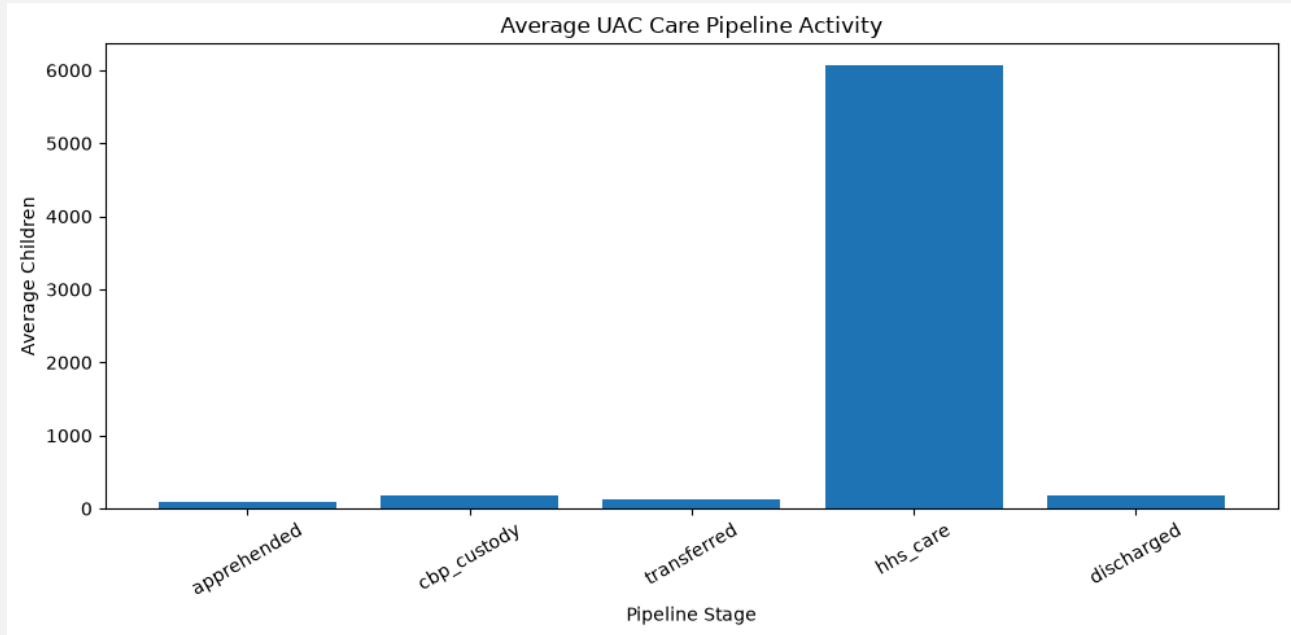


Figure 2. Aggregate UAC Care Pipeline Activity

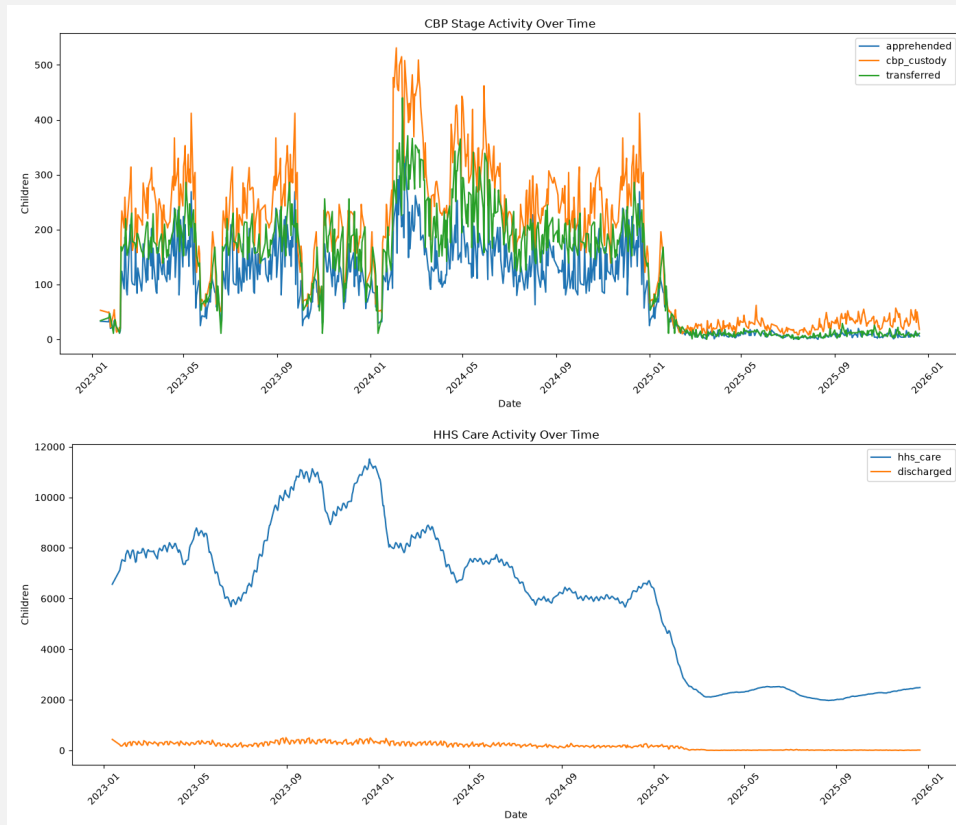


Figure 3. CBP and HHS Population Trends

Insert the CBP/HHS population time-series chart using separate scales or panels.

8. Efficiency and Outcome Analysis

8.1 Transfer Efficiency

The aggregate transfer efficiency ratio is 75.03%. This compares summed transfer activity with the summed reported CBP custody population. It is useful for period comparison, but does not measure how quickly an individual child was transferred.

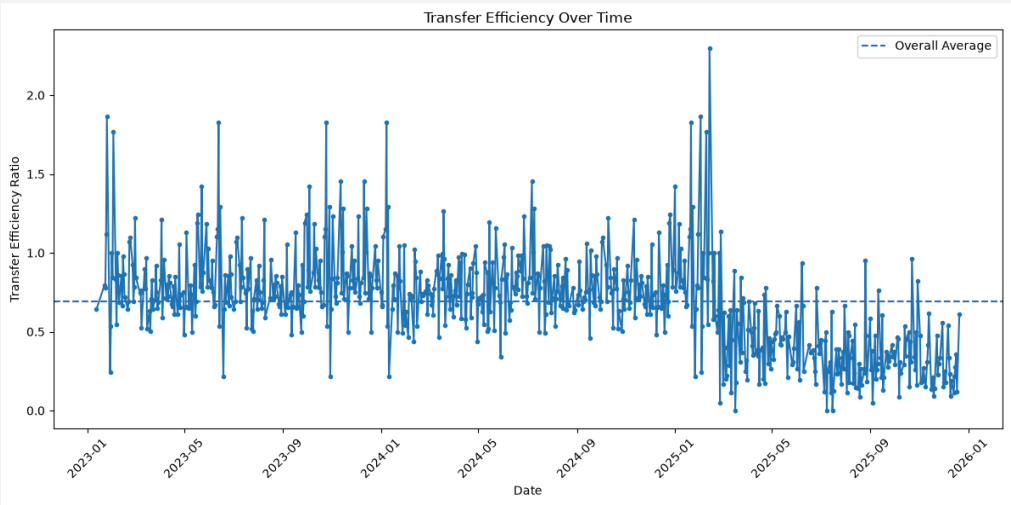


Figure 4. Transfer Efficiency Over Time

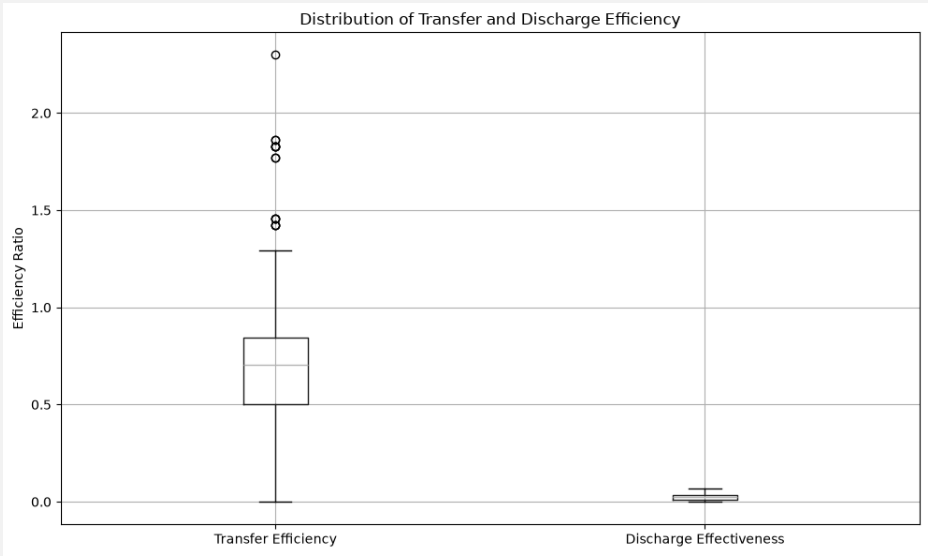
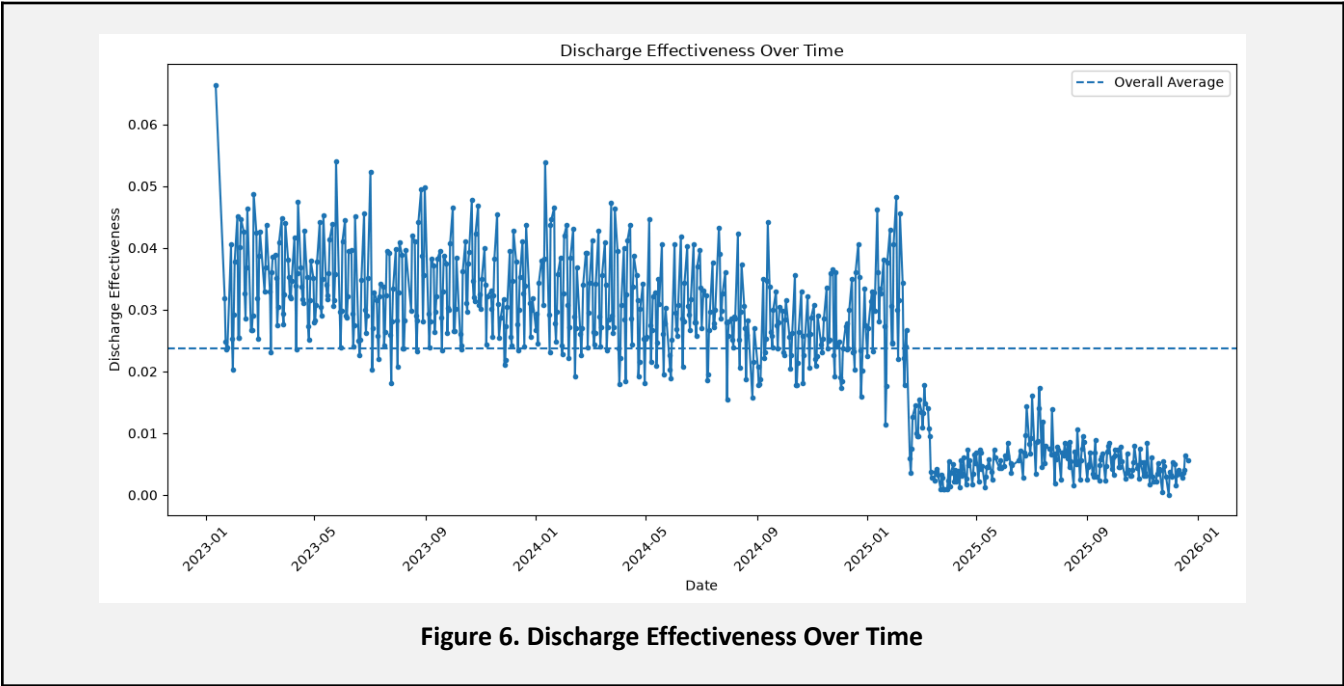


Figure 5. Transfer and Discharge Efficiency Distribution

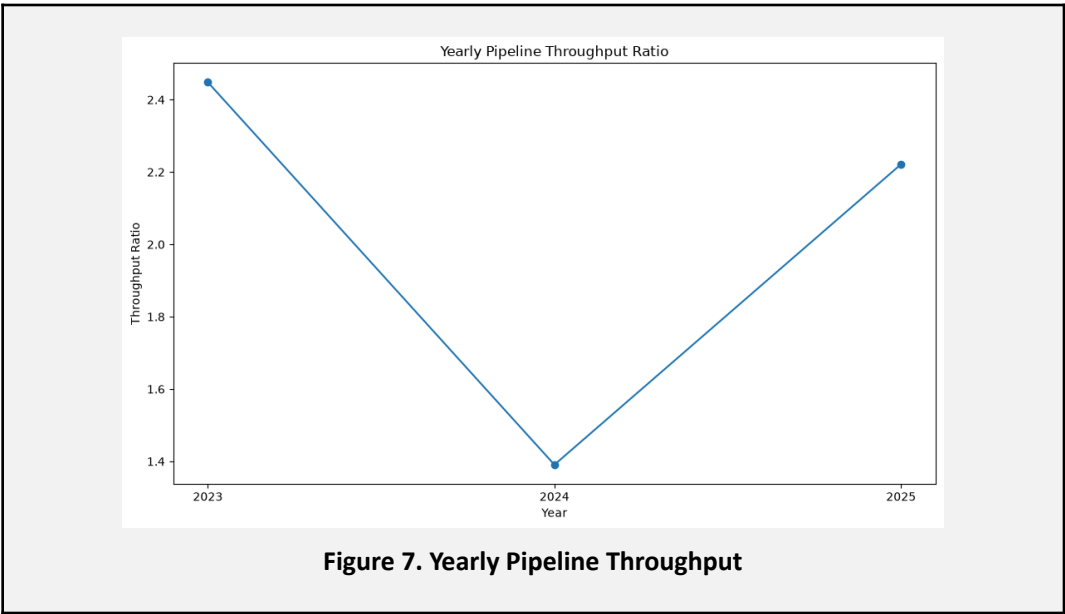
8.2 Discharge Effectiveness

The aggregate discharge effectiveness ratio is 2.86%. It represents discharge activity relative to the summed HHS care population and should not be described as an individual reunification success rate.



8.3 Pipeline Throughput

Aggregate pipeline throughput is 185.42%, calculated as total discharges divided by total apprehensions. This provides a broad activity comparison rather than a cohort-level completion rate.



9. Bottleneck and Operational Pressure Analysis

Mean CBP net pressure is -35.14 children per reporting observation, while mean HHS net pressure is -44.74. Positive pressure indicates that incoming activity exceeded the corresponding outgoing activity during an observation period. These measures are treated as pressure indicators rather than literal backlog counts.

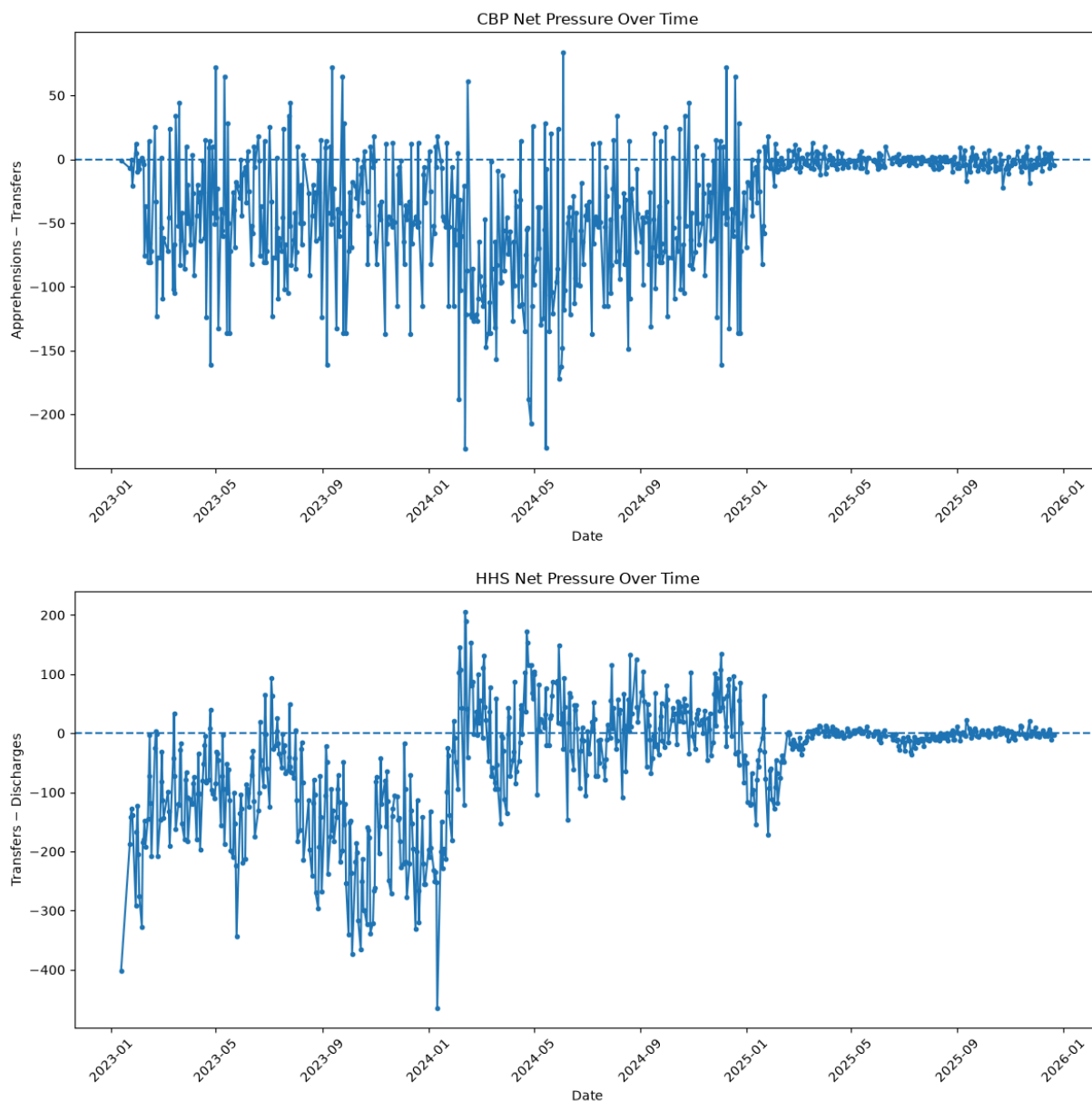


Figure 8. Operational Pressure Over Time

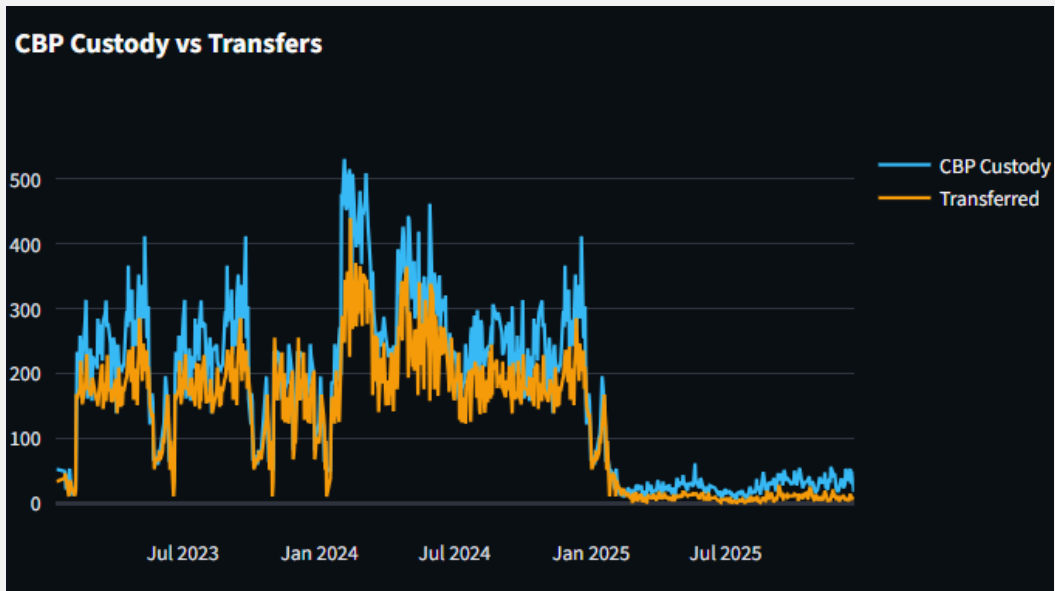


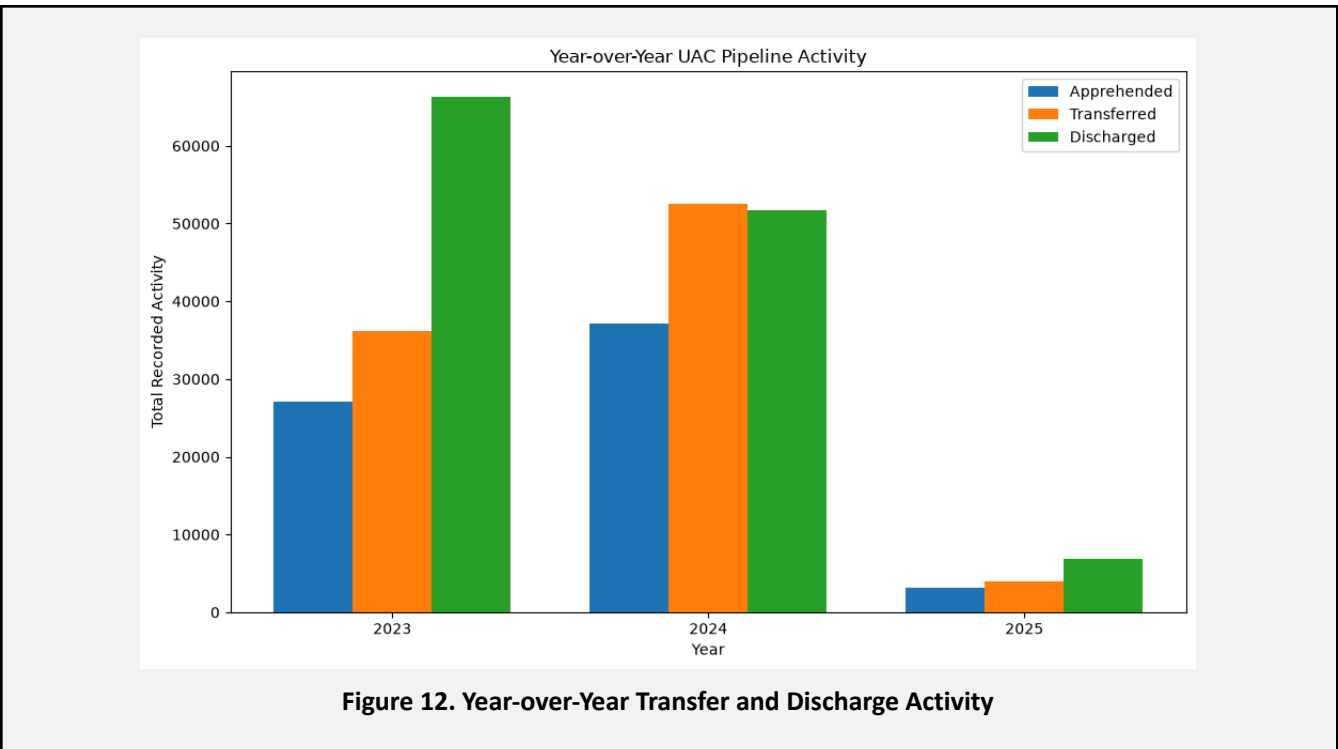
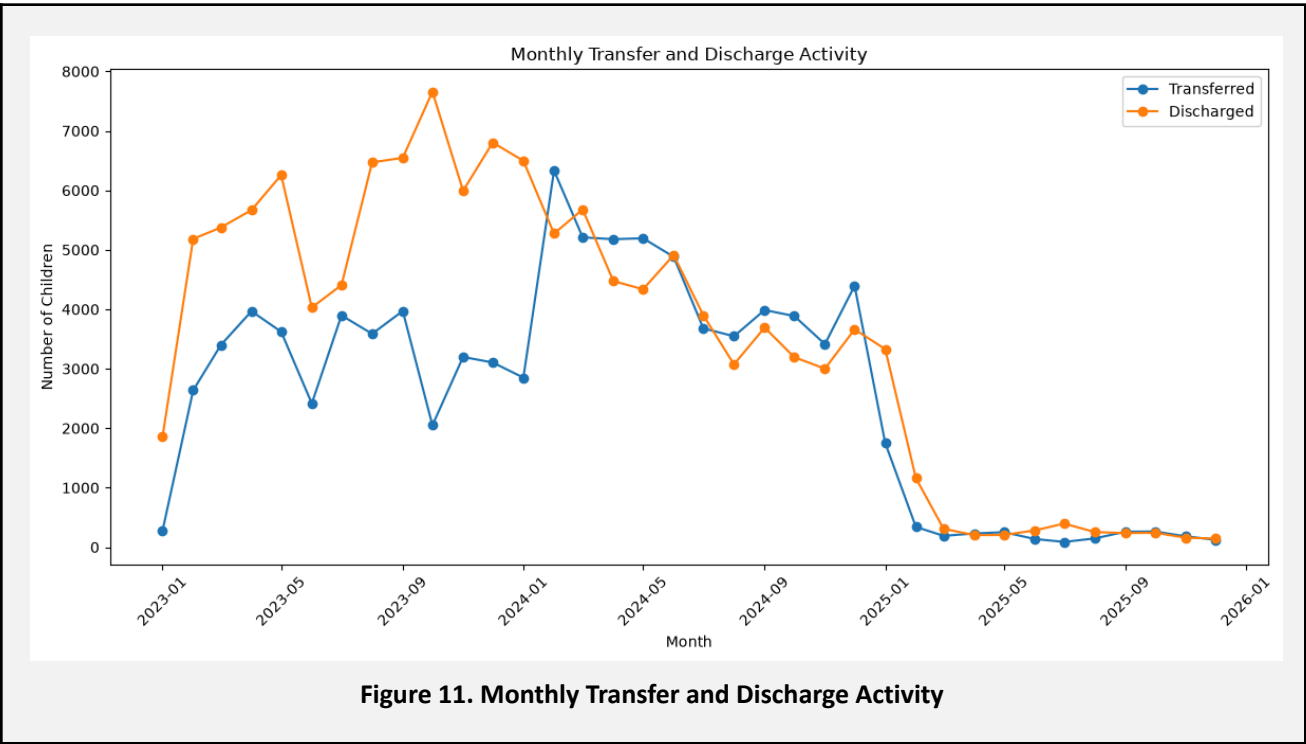
Figure 9. Transfer Efficiency vs CBP Pressure

Highest CBP Pressure Periods				
	date	apprehended	transferred	cbp_net_pressure
338	2024-06-04	260.0	176.0	84.0
67	2023-05-01	224.0	152.0	72.0
156	2023-09-12	224.0	152.0	72.0
465	2024-12-09	224.0	152.0	72.0
75	2023-05-11	269.0	204.0	65.0
164	2023-09-24	269.0	204.0	65.0
473	2024-12-19	269.0	204.0	65.0
262	2024-02-15	333.0	272.0	61.0
37	2023-03-20	202.0	158.0	44.0
126	2023-07-26	202.0	158.0	44.0
Highest HHS Pressure Periods				
	date	transferred	discharged	hhs_net_pressure
259	2024-02-12	440.0	234.0	206.0
260	2024-02-13	349.0	160.0	189.0
309	2024-04-23	307.0	134.0	173.0
263	2024-02-19	371.0	218.0	153.0
310	2024-04-24	307.0	154.0	153.0
335	2024-05-30	339.0	190.0	149.0
255	2024-02-06	323.0	178.0	145.0
461	2024-12-03	242.0	107.0	135.0
391	2024-08-20	245.0	112.0	133.0
274	2024-03-05	345.0	214.0	131.0

Figure 10. Highest-Pressure Reporting Periods

10. Temporal and Outcome Analysis

Monthly and yearly aggregation was used to examine longer-term activity patterns. Weekday analysis was interpreted cautiously because reporting coverage is uneven across weekdays.



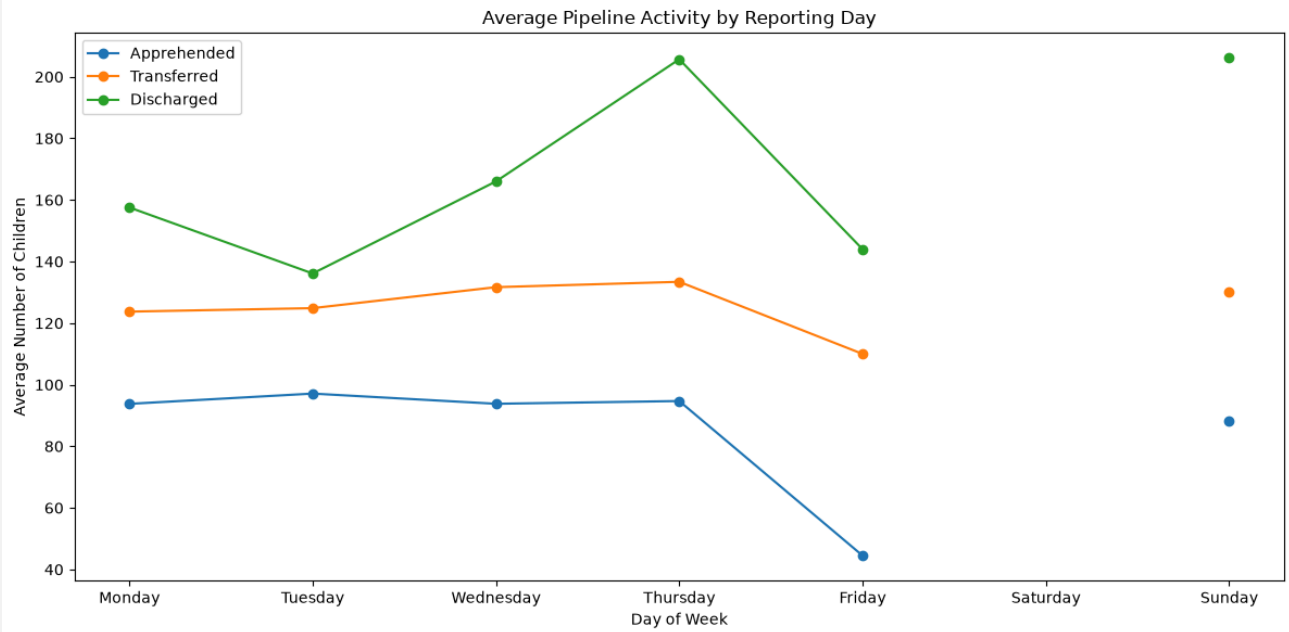


Figure 13. Weekday Operational Activity

11. Key Findings

- The cleaned analytical dataset contains 720 valid observations after removal of 450 blank rows.
- Aggregate reported activity includes 67,337 apprehensions, 92,641 transfers, and 124,853 discharges.
- The aggregate transfer efficiency ratio is 75.03%.
- The aggregate discharge effectiveness ratio is 2.86%.
- Aggregate pipeline throughput is 185.42%.
- Operational pressure varies across reporting periods, supporting threshold-based monitoring.
- The irregular reporting schedule limits direct interpretation of weekday comparisons.

12. Analytical Dashboard

A Streamlit dashboard was developed to operationalize the analysis. It presents KPI cards, interactive trend charts, transfer and discharge views, bottleneck analysis, temporal comparisons, insights, filters, and filtered-data export.

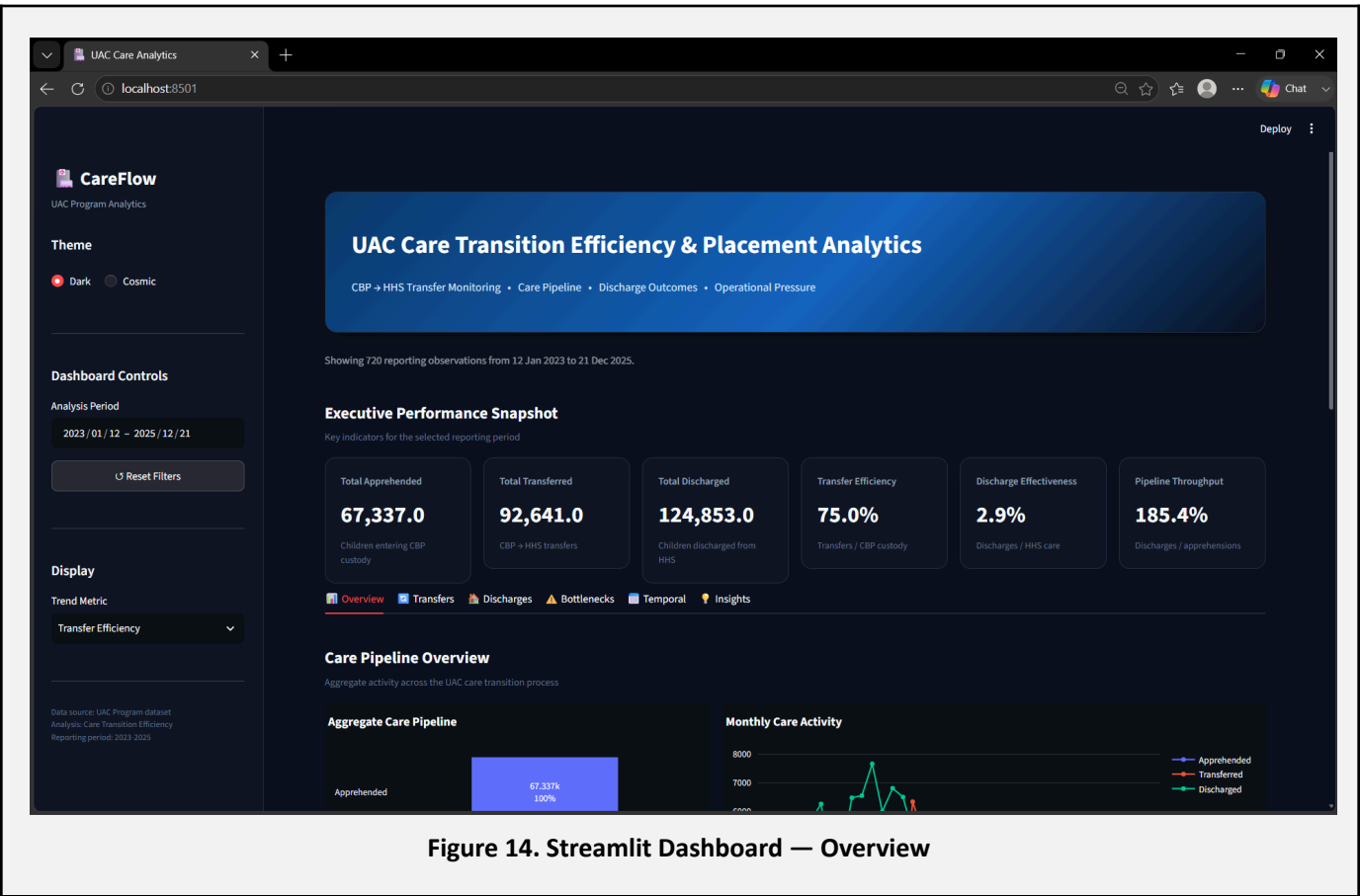


Figure 14. Streamlit Dashboard — Overview

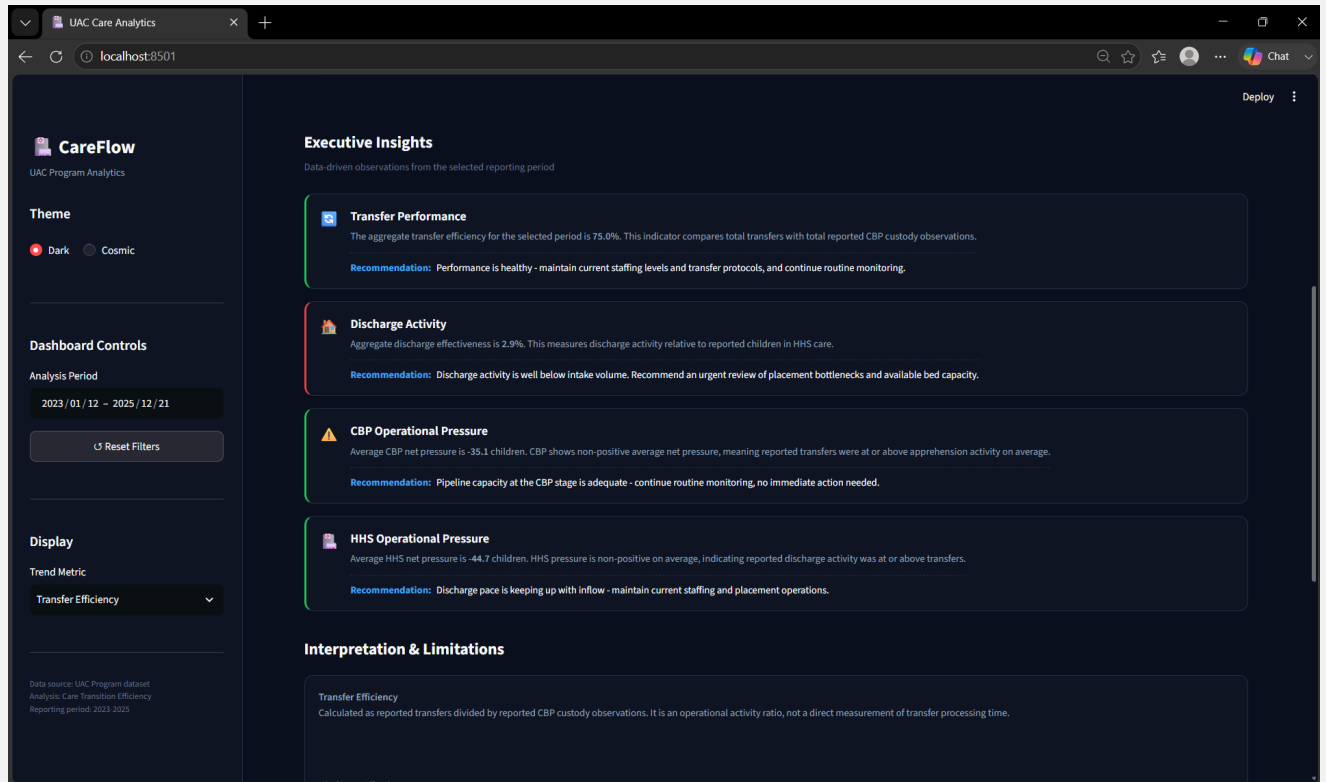


Figure 15. Streamlit Dashboard — Insights

13. Recommendations

Building on the findings above, this section sets out concrete recommendations for adopting and extending the care-transition analytics framework. The recommendations are organized around three areas: monitoring practice, data infrastructure, and governance of how the indicators are used and communicated.

13.1 Strengthen Operational Monitoring

- Adopt the transfer efficiency, discharge effectiveness, and pipeline throughput ratios as standing indicators alongside existing custody-population reporting, rather than as a one-time analytical exercise.
- Establish pressure thresholds for CBP Net Pressure and HHS Net Pressure so that sustained positive pressure — incoming activity outpacing outgoing activity — triggers a defined operational review rather than being noticed only retrospectively.
- Use the Outcome Stability (coefficient of variation) measure to distinguish reporting periods with genuinely volatile conditions from periods that are simply high or low in absolute terms.

13.2 Improve the Underlying Data Infrastructure

- Prioritize collection of individual-level, case-linked timestamps for apprehension, transfer, and discharge events, which would allow true elapsed-time measures to replace the current period-level ratios.
- Investigate and, where possible, correct the reporting gaps that produced 450 blank rows in the raw dataset, and standardize the reporting cadence to reduce the current one- to four-day irregularity.
- Add facility-level and geographic identifiers to the reporting structure so that bottlenecks can be localized rather than only observed in the aggregate.

13.3 Communicate Findings Responsibly

- When reporting these KPIs to operational or policy audiences, retain the framing used throughout this study: the ratios are comparative operational indicators, not measures of individual processing time or placement outcomes.
- Present efficiency and pressure trends together with the underlying volume figures (apprehensions, transfers, discharges) so that a ratio change is not read in isolation from the scale of activity producing it.
- Treat the weekday activity findings as provisional given uneven reporting coverage, and avoid using them as the sole basis for staffing or scheduling decisions until coverage is more consistent.

13.4 Sustain and Extend the Dashboard

- Maintain the Streamlit dashboard as a living operational tool, with a defined process for refreshing the underlying data on a regular cadence.
- Incorporate automated alerts for unusually high operational pressure, as identified in the Future Work section, so that threshold breaches are surfaced proactively rather than discovered during periodic review.

- Periodically validate the KPI definitions and thresholds with domain experts and official reporting documentation to ensure the indicators remain aligned with how the underlying programs actually operate.

14. Limitations

- The data are aggregate reporting-period observations and do not contain individual case histories.
- The efficiency ratios do not directly measure elapsed transfer or discharge time.
- Discharge activity is not equivalent to individual reunification success unless established by source definitions.
- The reporting schedule is irregular; missing dates were not treated as zero activity.
- Cumulative pressure does not necessarily equal a literal backlog.
- Weekday comparisons are affected by uneven reporting coverage.
- The analysis is descriptive and cannot establish causal explanations.

15. Future Work

- Add individual-level timestamps to calculate true turnaround times.
- Add geographic and facility-level variables for localized bottleneck analysis.
- Integrate staffing, capacity, and case-complexity variables.
- Develop forecasting and anomaly-detection models.
- Add automated dashboard alerts for unusually high operational pressure.
- Validate KPI definitions with domain experts and official reporting documentation.

16. Conclusion

This study set out to determine whether an aggregate, publicly reported dataset on the Unaccompanied Children care system could be restructured into a meaningful operational analytics framework, rather than serving only as a static record of custody and care populations. The evidence developed across Sections 6 through 12 supports an affirmative answer: by separating stock measures from flow measures and pairing them into ratio-based indicators — transfer efficiency, discharge effectiveness, pipeline throughput, CBP and HHS net pressure, and outcome stability — the analysis produces a process-oriented view of the pipeline that a raw population count cannot provide on its own.

Applied to 720 valid observations spanning January 2023 to December 2025, the framework found an aggregate transfer efficiency of 75.03%, an aggregate discharge effectiveness of 2.86%, and an aggregate pipeline throughput of 185.42%, alongside average CBP and HHS net pressure figures of -35.14 and -44.74 respectively. Beyond these headline figures, the monthly, yearly, and weekday breakdowns identify specific periods of elevated operational pressure that would not be visible from custody-population counts alone, and the resulting Streamlit dashboard translates the framework into a tool that can be used for ongoing monitoring rather than a single retrospective report.

At the same time, the study has been deliberate about the boundaries of what this kind of aggregate, reporting-period data can support. The KPIs developed here are comparative operational indicators, not direct measures of individual processing time, individual placement success, or literal backlog size, and the analysis throughout has avoided causal claims that the data cannot substantiate. This distinction is not a weakness of the framework so much as a necessary discipline: the value of the indicators lies precisely in their consistent, reproducible construction, which depends on their being interpreted for what they are.

Looking ahead, the framework developed here is best understood as a foundation rather than a finished system. With the addition of individual case-level timestamps, geographic and facility identifiers, staffing and capacity data, and case-complexity variables — as outlined in the Recommendations and Future Work sections — the same underlying approach could be extended toward true turnaround-time measurement, predictive and anomaly-detection modeling, and facility-level bottleneck detection. In the meantime, the KPI and dashboard framework presented in this paper offers a practical, immediately usable step toward monitoring the UAC care transition pipeline as a system of connected transitions rather than a series of disconnected counts.

Appendix A — Supporting Figures

This appendix consolidates the full set of figures referenced throughout the paper, reproduced here at larger size for closer inspection.

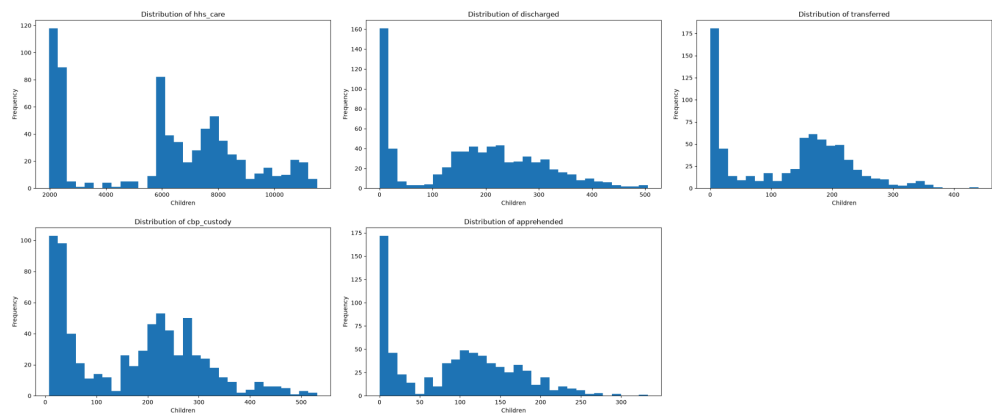


Figure A1. Distribution of Major Pipeline Variables

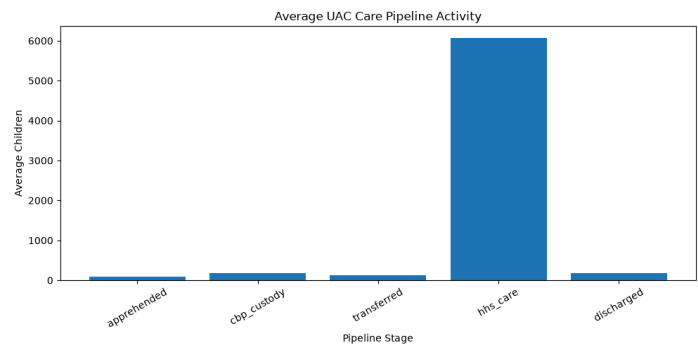


Figure A2. Aggregate UAC Care Pipeline Activity

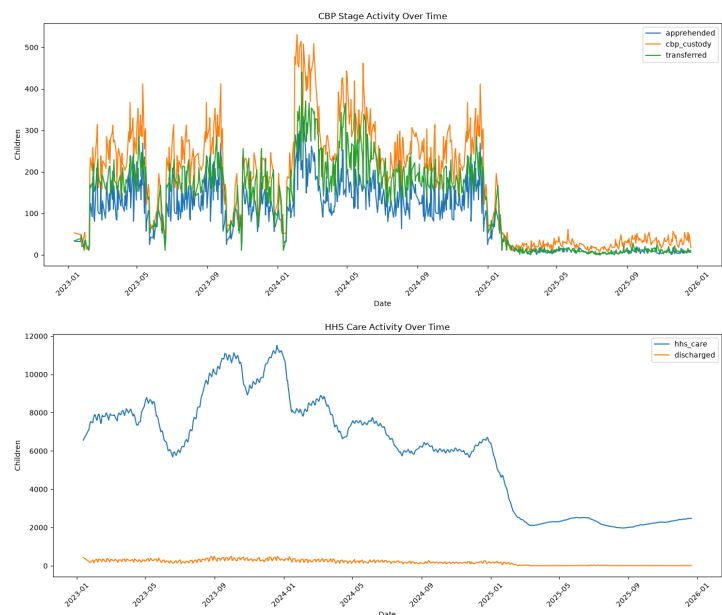


Figure A3. CBP and HHS Population Trends

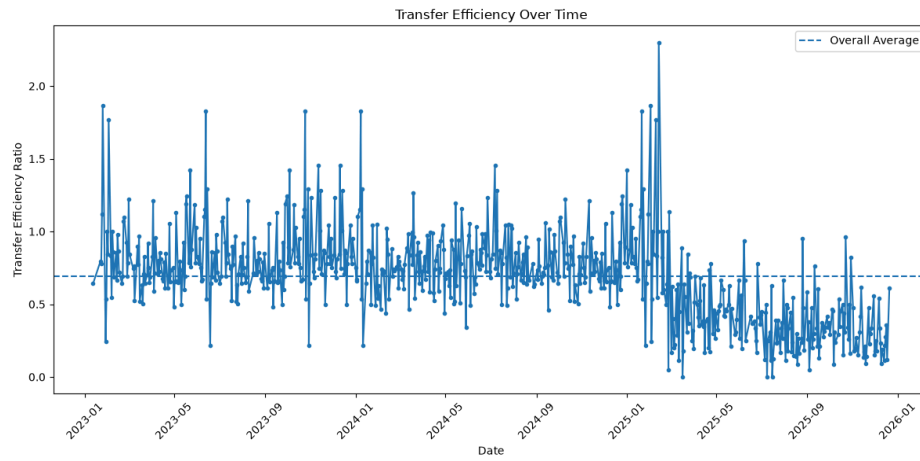


Figure A4. Transfer Efficiency Over Time

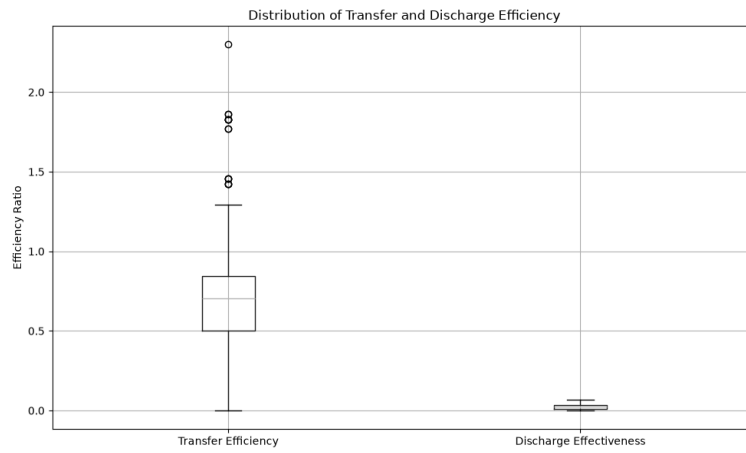


Figure A5. Transfer and Discharge Efficiency Distribution

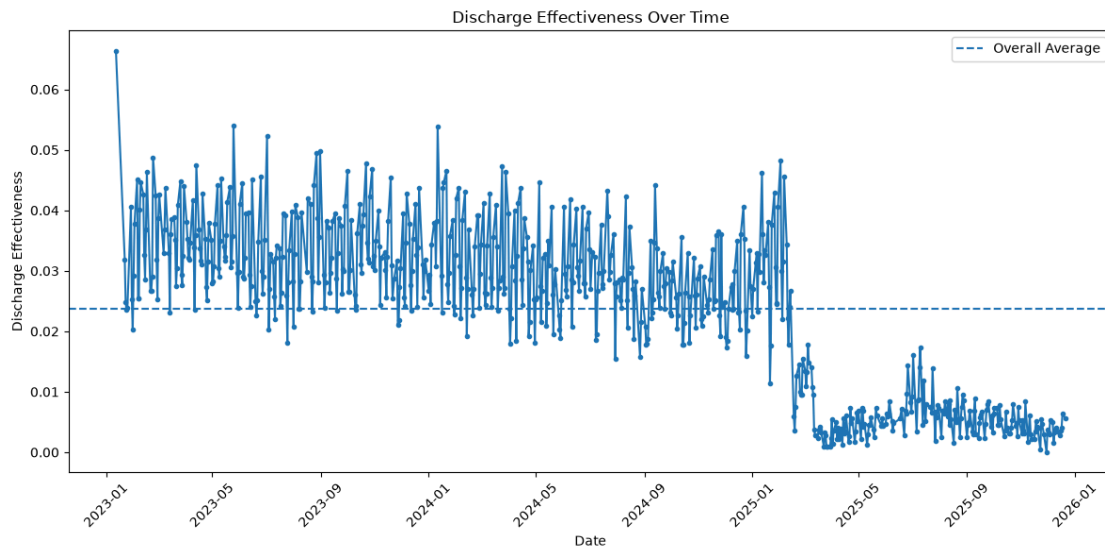


Figure A6. Discharge Effectiveness Over Time

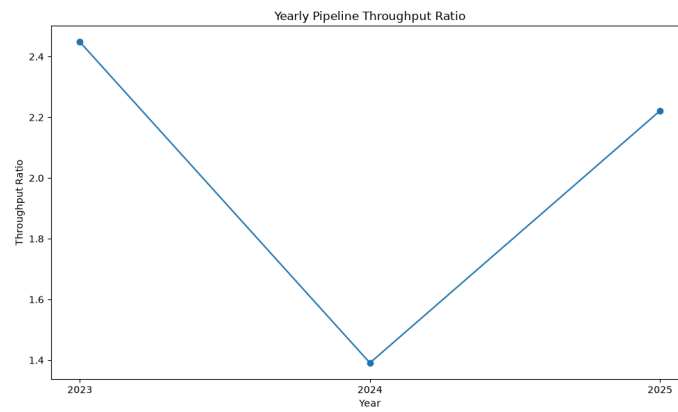


Figure A7. Yearly Pipeline Throughput

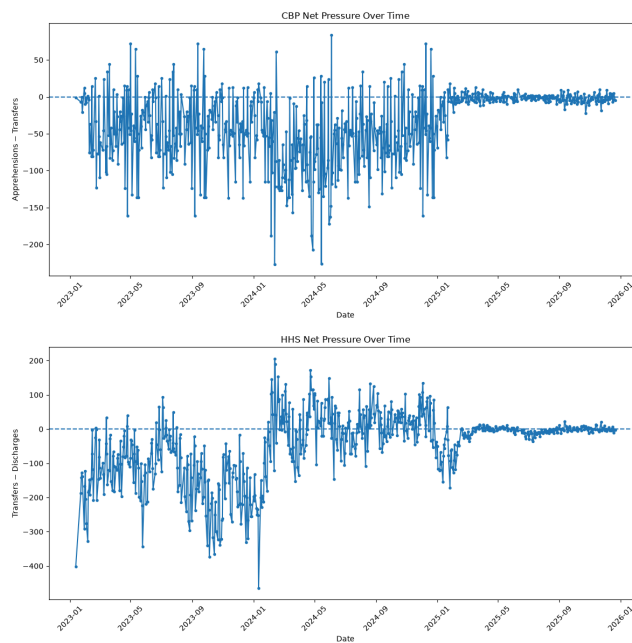


Figure A8. Operational Pressure Over Time

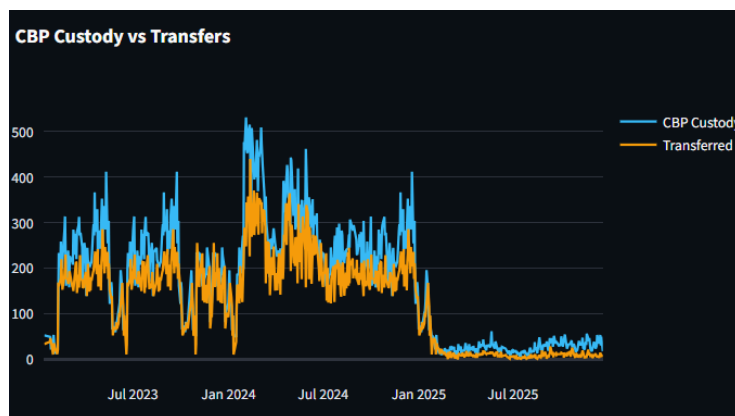


Figure A9. Transfer Efficiency vs CBP Pressure

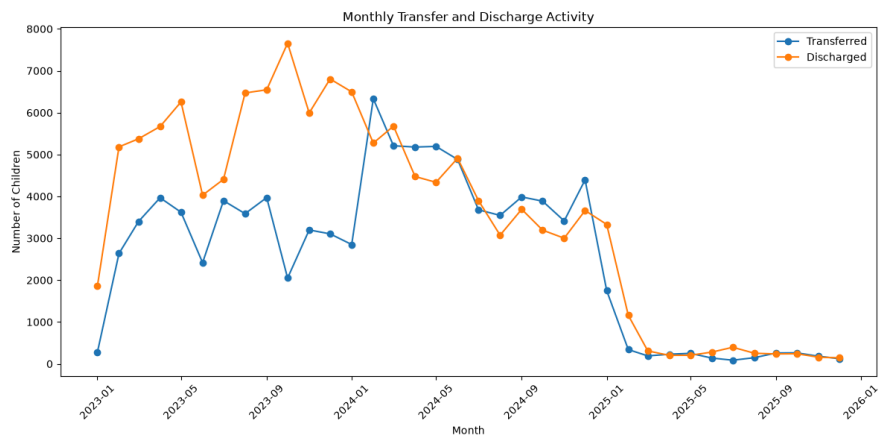


Figure A10. Monthly Transfer and Discharge Activity

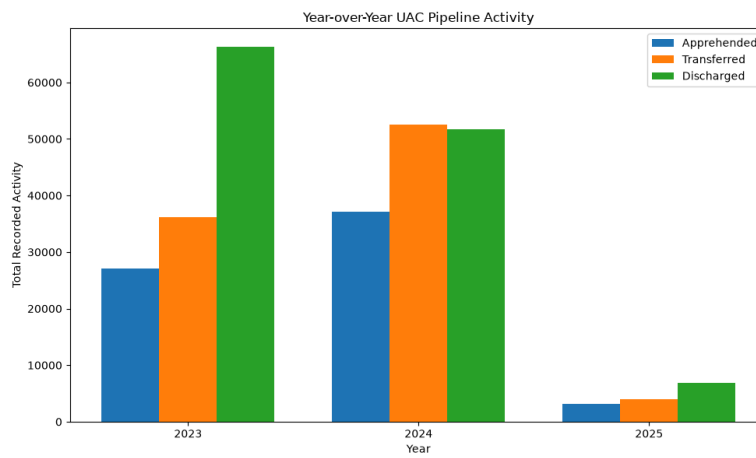


Figure A11. Year-over-Year Transfer and Discharge Activity

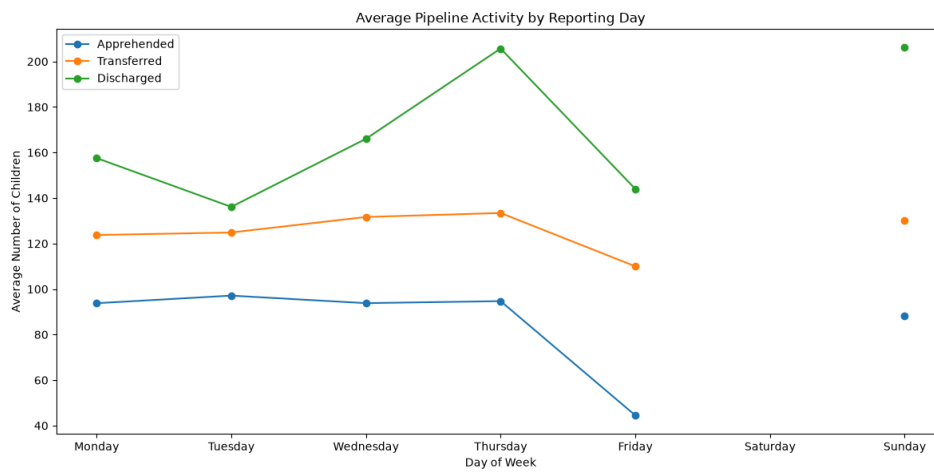


Figure A12. Weekday Operational Activity

Appendix B — Supporting Analytical Tables

B.1 KPI Summary

KPI	Value
Total apprehended	67,337
Total transferred	92,641
Total discharged	124,853
Transfer efficiency	75.03%
Discharge effectiveness	2.86%
Pipeline throughput	185.42%
Average CBP net pressure	-35.14
Average HHS net pressure	-44.74

B.2 Highest CBP Pressure Periods

Reporting periods with the largest positive gap between apprehensions and transfers (CBP Net Pressure = Apprehended – Transferred).

Date	Apprehended	Transferred	CBP Net Pressure
2024-06-04	260.0	176.0	84.0
2023-05-01	224.0	152.0	72.0
2023-09-12	224.0	152.0	72.0
2024-12-09	224.0	152.0	72.0
2023-05-11	269.0	204.0	65.0
2023-09-24	269.0	204.0	65.0
2024-12-19	269.0	204.0	65.0
2024-02-15	333.0	272.0	61.0
2023-03-20	202.0	158.0	44.0
2023-07-26	202.0	158.0	44.0

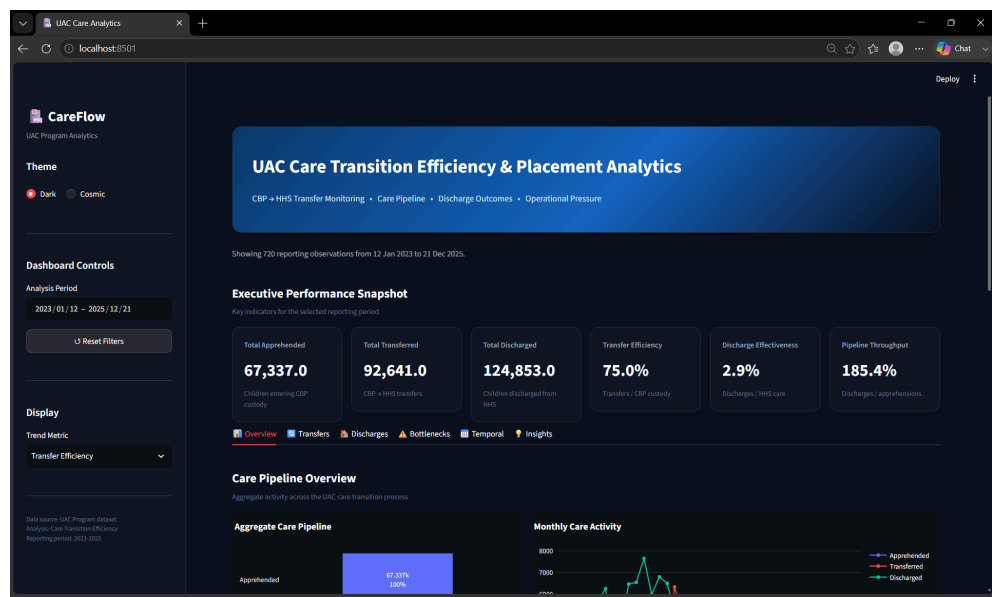
B.3 Highest HHS Pressure Periods

Reporting periods with the largest positive gap between transfers and discharges (HHS Net Pressure = Transferred – Discharged).

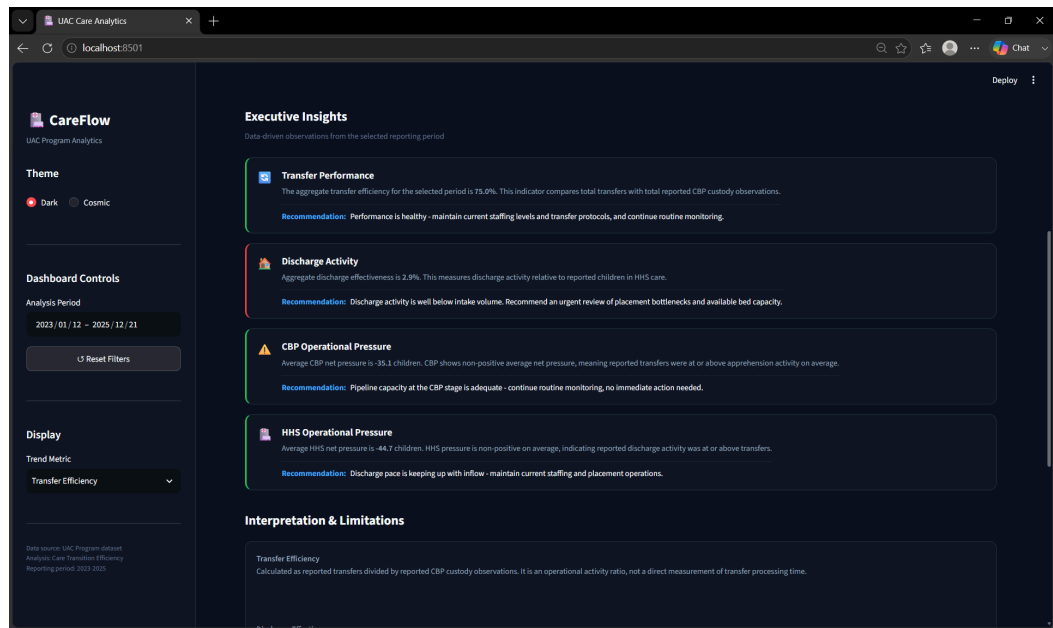
Date	Transferred	Discharged	HHS Net Pressure
2024-02-12	440.0	234.0	206.0
2024-02-13	349.0	160.0	189.0
2024-04-23	307.0	134.0	173.0
2024-02-19	371.0	218.0	153.0
2024-04-24	307.0	154.0	153.0
2024-05-30	339.0	190.0	149.0
2024-02-06	323.0	178.0	145.0
2024-12-03	242.0	107.0	135.0
2024-08-20	245.0	112.0	133.0
2024-03-05	345.0	214.0	131.0

Appendix C — Project Artifacts

This appendix documents the interactive analytical dashboard developed to operationalize the KPI framework described in Section 12. The dashboard was built in Streamlit and provides an overview panel with executive KPI cards, transfer and discharge trend views, bottleneck analysis, temporal comparisons, and a filtered data export function covering the full 720-observation, 12 Jan 2023 – 21 Dec 2025 reporting window.



Artifact C1. Streamlit Dashboard — Overview panel with KPI cards and pipeline trend charts



Artifact C2. Streamlit Dashboard — Executive Insights panel with narrative interpretation and recommendations

Both panels draw on the same cleaned 720-observation dataset and KPI definitions documented in Section 5.3, so the figures shown on the dashboard match the aggregate values reported throughout this paper.

Appendix D — References

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